PARISHRAM 2.0

CLASS 12TH 2026

MATRIX

MATHS | LECTURE-1

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TOPICS TO BE COVERED

1. What are Matrices?
2. Order of a Matrix ??
3. Addition & Subtraction of Matrices

Why Fear When Rolex Sir is Here



What Are Matrices?

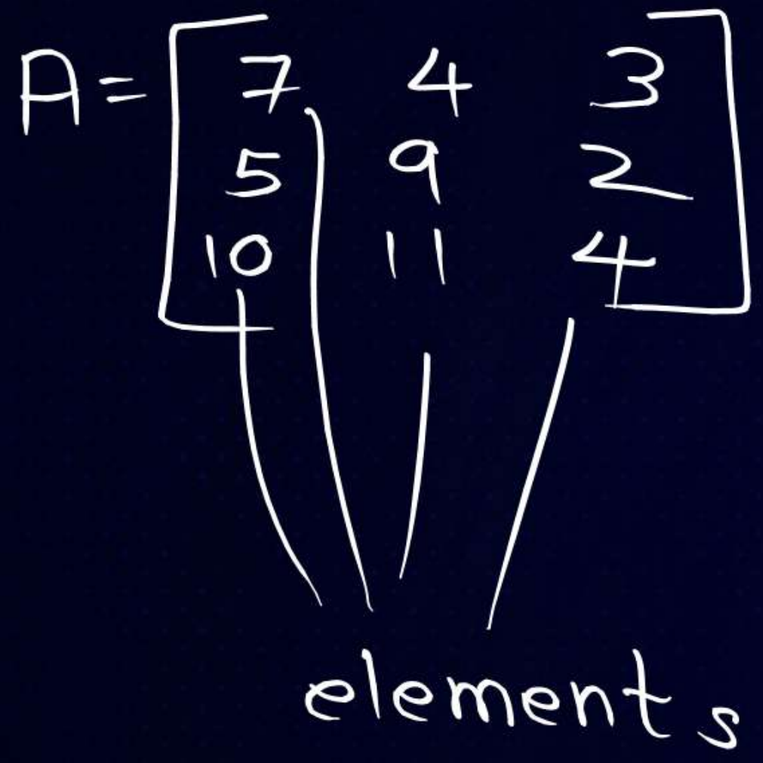
A matrix is a rectangular arrangement of numbers or functions in rows and columns, enclosed within brackets.

Each number or element in a matrix is called an element or entry of the matrix.



Rows → sleeping

Coloumns → standing



→ Matrix

	C_1	C_2	C_3	
$B =$	10	4	3	R_1
	7	9	2	R_2

Matrices In Real Life

Excel Sheet

- **Tabular Form**

	Math	Science	English
Aman	85	90	78
Bhavya	76	88	84
Chirag	92	81	79

- **Matrix Form**

Math Sci Eng

$$M = \begin{bmatrix} 85 & 90 & 78 \\ 76 & 88 & 84 \\ 92 & 81 & 79 \end{bmatrix}$$

Aman
Bhavya
Chirag

Order of a Matrix

The order of a matrix refers to its dimensions–that is, the number of rows and columns it has. It is usually written as:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 9 & 10 & 13 \end{bmatrix} \begin{matrix} \rightarrow R_1 \\ \rightarrow R_2 \end{matrix}$$

$C_1 \quad C_2 \quad C_3$

No of Rows = 2

No of Coloumns = 3

Order = No of rows × No of coloumns

Order = 2 × 3

$$C = \begin{bmatrix} 2 & 9 \\ 4 & 2 \\ 1 & 7 \end{bmatrix} \begin{matrix} \rightarrow R_1 \\ \rightarrow R_2 \\ \rightarrow R_3 \end{matrix}$$

$C_1 \quad C_2$

Order of Matrix C

↓

3 × 2

Types of Matrices

1. Row Matrix

A matrix with only one row.

$$A = \begin{bmatrix} 1 & 7 & 9 \end{bmatrix}_{R_1}$$

$$B = \begin{bmatrix} 7 & 15 \end{bmatrix}_{R_1}$$

2. Column Matrix

A matrix with only one column.

$$C = \begin{bmatrix} 7 \\ 9 \\ 15 \end{bmatrix}_{C_1}$$

$$D = \begin{bmatrix} 7 \\ 19 \end{bmatrix}_{C_1}$$

Types of Matrices

3. Square Matrix

Number of rows = Number of columns

$$B = \begin{bmatrix} 2 & 4 \\ 9 & 13 \end{bmatrix} \begin{matrix} \rightarrow R_1 \\ \rightarrow R_2 \end{matrix}$$

$C_1 \quad C_2$

2×2

$$C = \begin{bmatrix} 17 & 9 & 14 \\ 10 & 4 & 12 \\ 9 & 2 & 7 \end{bmatrix}$$

3×3

4. Rectangular Matrix

Number of rows ≠ Number of columns

$$X = \begin{bmatrix} 7 & 9 & 4 \\ 2 & 19 & 10 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \end{matrix}$$

$C_1 \quad C_2 \quad C_3$

2×3

No of rows ≠ No of coloumns

Types of Matrices

5. Diagonal Matrix

A square matrix where all non diagonal elements are zero.

$$Y = \begin{bmatrix} 7 & 0 \\ 0 & 4 \end{bmatrix}$$

Diagonal elements

$$Z = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 10 \end{bmatrix}$$

6. Scalar Matrix

A diagonal matrix where all diagonal elements are equal.

$$W = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

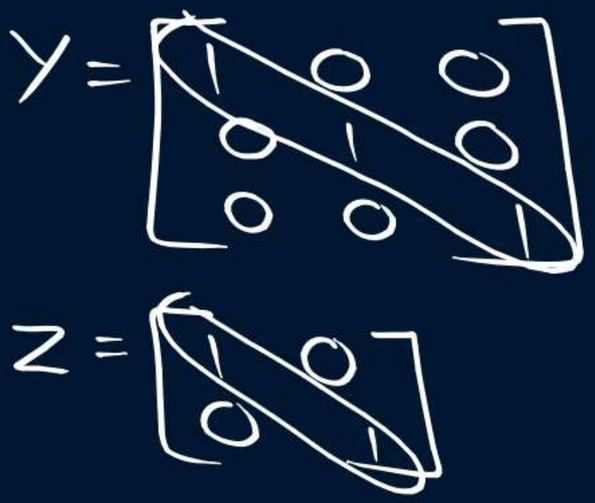
Diagonal vale elements
↳ same

$$A = \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix}$$

Types of Matrices

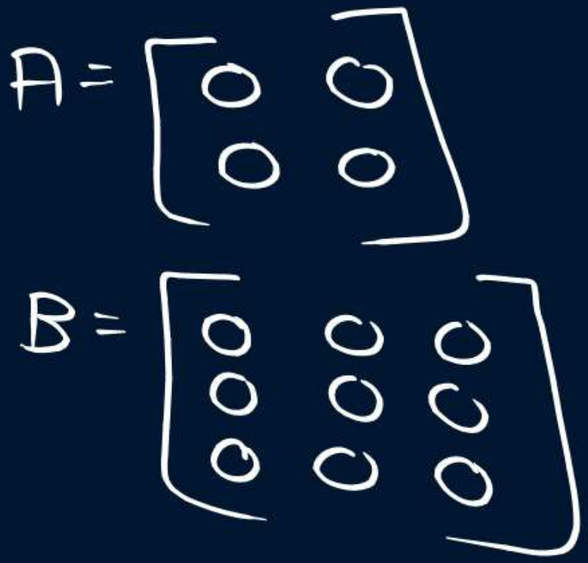
7. Identity Matrix (Unit Matrix)

A square matrix with 1s on the diagonal and 0s elsewhere.



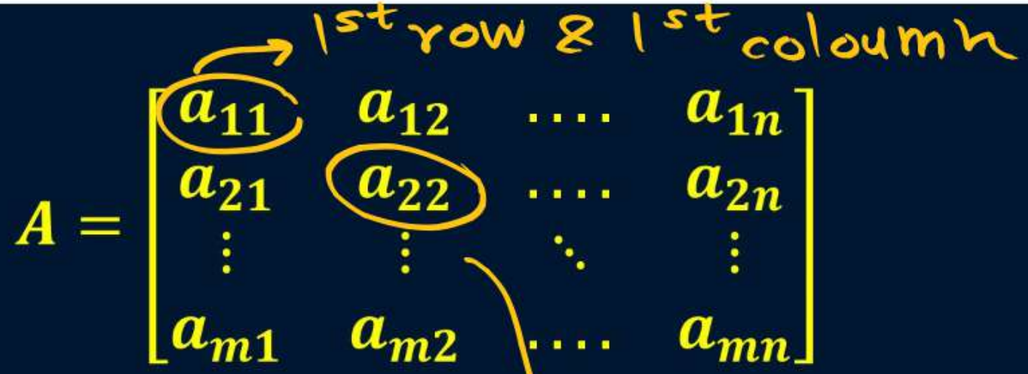
8. Zero Matrix (Null Matrix)

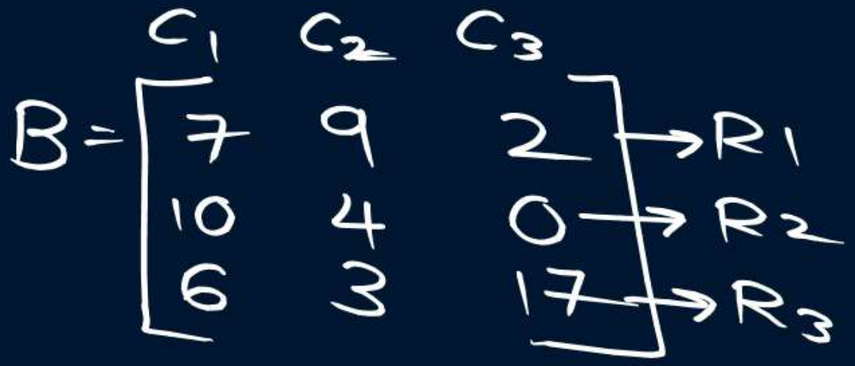
All elements are zero.



Representation of Matrix

A matrix is usually represented by a capital letter such as A.
If its has m rows and n columns, it is called an $m \times n$ matrix.





0 → position (b_{23})

9 → positi

10 → b_{21}

$$A = \begin{bmatrix} 9 & 33 & 17 \\ 4 & 5 & 13 \\ 73 & 44 & 3 \end{bmatrix}$$

Chat Box

↓

$a_{31} = 73$

$a_{23} = 13$

$a_{22} = 5$

$a_{12} = 33$

Equality of Matrices

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be equal, written as A = B.

1. **They have the same order (same number of rows and columns)**
2. **Their corresponding elements are equal.**

$$A = \begin{bmatrix} 2 & 4 \\ 5 & 7 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 4 \\ 5 & 7 \end{bmatrix}$$

A = B

$$A = \begin{bmatrix} 2 & 4 \\ 5 & 7 \end{bmatrix}_{2\times 2} \qquad B = \begin{bmatrix} 2 & 4 \\ 5 & 7 \end{bmatrix}_{2\times 2}$$

Poll Qt

1) Matrix A = Matrix B

$$A = \begin{bmatrix} 7 & 5 \\ x & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 7 & y \\ 9 & 2 \end{bmatrix}$$

$x = ?$ $y = ?$

a) $x = 2, y = 5$

b) $x = 9, y = 5$ ✓

c) $x = 5, y = 9$

d) $x = 7, y = 2$

$x = 9$

$y = 5$

QUESTION

$\begin{bmatrix} 2x+3 & y-1 \\ 4 & z+5 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 4 & 2z \end{bmatrix}$. Find values x, y and z.

$2x + 3 = 7$

$2x = 7 - 3$

$2x = 4$

$x = 4/2$

$\boxed{x = 2}$

$y - 1 = 3$

$y = 3 + 1$

$\boxed{y = 4}$

$z + 5 = 2z$

$5 = 2z - 1z$

$\boxed{5 = z}$



Addition and Subtraction of Matrices

Two matrices can be added or subtracted only if they have the same order (same number of rows and columns.)

The Operation is performed by adding or subtracting corresponding elements.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

2×2 2×2

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix}$$

$$\mathbf{A} - \mathbf{B} = \begin{bmatrix} a_{11} - b_{11} & a_{12} - b_{12} \\ a_{21} - b_{21} & a_{22} - b_{22} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 7 & 9 \\ 10 & 11 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 1+7 & 2+9 \\ 3+10 & 4+11 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 8 & 11 \\ 13 & 15 \end{bmatrix}$$

$$X = \begin{bmatrix} 77 & 12 & 9 \\ 14 & 13 & 9 \\ 10 & 4 & 8 \end{bmatrix} \quad Y = \begin{bmatrix} 10 & 2 & 4 \\ 9 & 8 & 15 \\ 10 & 3 & 9 \end{bmatrix}$$

$$X + Y = \begin{bmatrix} 77+10 & 12+2 & 9+4 \\ 14+9 & 13+8 & 9+15 \\ 10+10 & 4+3 & 8+9 \end{bmatrix}$$

$$X + Y = \begin{bmatrix} 87 & 14 & 13 \\ 23 & 21 & 24 \\ 20 & 7 & 17 \end{bmatrix}$$

Poll Qt

1) $A = \begin{bmatrix} 7 & 4 \\ 2 & 10 \end{bmatrix}$ $B = \begin{bmatrix} 9 & 2 \\ 7 & 4 \end{bmatrix}$

A + B

a) $\begin{bmatrix} 16 & 6 \\ 9 & 10 \end{bmatrix}$

b) $\begin{bmatrix} 16 & 6 \\ 9 & 14 \end{bmatrix}$

c) $\begin{bmatrix} 10 & 6 \\ 9 & 14 \end{bmatrix}$

d) $\begin{bmatrix} 10 & 16 \\ 9 & 14 \end{bmatrix}$

A - B

a) $\begin{bmatrix} -2 & 2 \\ -5 & 6 \end{bmatrix}$

b) $\begin{bmatrix} 4 & 2 \\ 5 & 6 \end{bmatrix}$

c) $\begin{bmatrix} 7 & 9 \\ 5 & 6 \end{bmatrix}$

d) $\begin{bmatrix} -2 & 2 \\ 10 & 6 \end{bmatrix}$

QUESTION

Find the matrix B.

$$A + B = \begin{bmatrix} 10 & 5 & 9 \\ 6 & 12 & 8 \\ 0 & 14 & 4 \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 7 & 4 \\ -3 & 6 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 & 1 \\ 1 & 7 & 4 \\ -3 & 6 & 2 \end{bmatrix} + B = \begin{bmatrix} 10 & 5 & 9 \\ 6 & 12 & 8 \\ 0 & 14 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 7 & 3 & 8 \\ 5 & 5 & 4 \\ 3 & 8 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 10 & 5 & 9 \\ 6 & 12 & 8 \\ 0 & 14 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 2 & 1 \\ 1 & 7 & 4 \\ -3 & 6 & 2 \end{bmatrix} = \begin{bmatrix} 10-3 & 5-2 & 9-1 \\ 6-1 & 12-7 & 8-4 \\ 0-(-3) & 14-6 & 4-2 \end{bmatrix}$$



THANK YOU